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authorized hydrographic office or other relevant government institution, as required by SOLAS regulations V/19 and V/27.

1.4 ECDIS should facilitate simple and reliable updating of the ENDS.

1.5 ECDIS should reduce the navigational workload compared to using the paper chart and paper nautical publications. It should enable the mariner to execute in a convenient and timely manner all route planning, route monitoring and positioning. It should be capable of continuously indicating, monitoring and recording the ship's position.

1.6 The ECDIS display may also b e used for the display of radar, radar tracked target information, AIS and other appropriate data layers to assist in route monitoring.

1.7 ECDIS should provide appropriate alerts or indications with respect to the information displayed or malfunction of the equipment (see appendix 5). ECDIS should meet the requirements of the Performance standards for bridge alert management (resolution MSC.302(87)).

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requirements of the Performance standards for bridge alert management (resolution MSC.302(87)).

1.8 When the relevant chart information is not available in the appropriate form (see section 4), some ECDIS equipment may operate in the raster chart display system (RCDS) mode as defined in appendix 7. RCDS mode of operation should conform to p erformance standards not inferior to those set out in appendix 7.

2 APPLICATION OF THESE STANDARDS

2.1 These performance standards should apply to all ECDIS equipment carried on all ships, as follows:

.1 dedicated stand-alone workstation; and

,2 a multifunction workstation as part of an INS.

2.2 These performance standards apply to ECDIS mode of operation, ECDIS in RCDS mode of operation as specified in appendix 7 and ECDIS backup arrangements as specified in appendix 6.

RESOLUTION MSC.530(106) (adopted on 7 November 2022)
PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS)

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2.3 Requirements for structure, format, encryption presentation of the ENDS are within the scope of relevant IHO standards, including those listed in appendix 1.

2.4 In addition to the general requirements set out in resolution A.694(17) 1 and the presentation requirements set out in resolution MSC.191(79), as amended, ECDIS equipment should meet the requirements of these standards and follow the relevant guidelines on ergonomic principles adopted by the Organization.2

3 DEFINITIONS

For the purpose of these performance standards:

3.1 Electronic chart display and information system (ECDIS) means a navigation information system which with adequate backup arrangements can be accepted as complying
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information system which with adequate backup arrangements can be accepted as complying with the up -to-date nautical chart and nautical publications required by SOLAS regulations V/19 and V/27, by displaying selected information from a system database with positional information from navigation sensors to assist the mariner in route planning and route monitoring and, if required, display additional navigation-related information.

3.2 Electronic navigational chart (ENC) means the database, standardized as to content, structure and format, issued for use with ECDIS by or on the authority of a government, authorized hydrographic office or other relevant government institution, and conforming to IHO standards. The ENC contains all the nautical chart information necessary for safe navigation.

3.3 Electronic navigational data service (ENDS) means a special -purpose database compiled from nautical chart and nautical publication data, standardized as to content,
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compiled from nautical chart and nautical publication data, standardized as to content, structure and format, issued for use with ECDIS by or on the authority of a government, authorized hydrographic office or other relevan t government institution, and conforming to IHO standards; and, which is designed to meet the requirement of marine navigation and the nautical charts and nautical publications carriage requirements in SOLAS regulations V/19 and V/27. The navigational base layer of ENDS is the electronic navigational chart (ENC).

3.4 System database means a database, in the manufacturer's internal ECDIS format, resulting from the lossless transformation of the ENDS contents and its updates. It is this

database that i s accessed by ECDIS for the display generation and other navigational functions, and is equivalent to up-to-date ENDS.

3.5 Standard display is the display mode intended to be used as a minimum during route

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functions, and is equivalent to up-to-date ENDS.

3.5 Standard display is the display mode intended to be used as a minimum during route
planning and route monitoring. The chart content is listed in appendix 2.

3.6 Display base means the chart content as listed in appendix 2 and which cannot be removed from the display. It is not intended to be sufficient for safe navigation.

3.7 Further information on ECDIS definitions may be found in IHO Hydrographic Dictionary Publication S-32 (see appendix 1).

1 MSC/Circ.982.

2 Refer to Publication IEC 60945.

MODULE A DATABASE

4 PROVISION AND UPDATING

4.1 The ENDS information to be used in ECDIS should be issued by or on the authority

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MODULE A DATABASE

4 PROVISION AND UPDATING

4.1 The ENDS information to be used in ECDIS should be issued by or on the authority of a government, government -authorized hydrographic office or other relevant government institution, and conform to IHO standards as listed in appendix 1.

4.2 The contents of the system database should be adequate and up to date for the intended voyage to comply with SOLAS regulations V/19 and V/27.

4.3 It should not be possible to alter the contents of the ENDS or system database information transformed from the ENDS. The display of the content of ENDS should be compliant with IHO standards including rules set for interoperability.

4.4 ECDIS should be capable of accepting official updates to the ENDS provided in conformity with IHO standards. These updates should be automatically applied to the system database. By whatever means updates are received, the implementation procedure should not interfere with the display in use.

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database. By whatever means updates are received, the implementation procedure should not interfere with the display in use.

4.5 ECDIS should also be capable of accepting updates to the ENDS data entered manually with simple means for verification prior to the final acceptance of the data. They should be distinguishable on the display from ENDS i nformation and its official updates and not affect display legibility.

4.6 ECDIS should keep and display on demand a record of updates including time of application to the system database. This record should include updates for each ENDS until it is superseded by a new edition.

4.7 ECDIS should allow the mariner to display updates in order to review their contents and to ascertain that they have been included in the system database.

4.8 ECDIS should be capable of accepting ENDS in accordance with the I HO Data Protection Scheme.3

MODULE B OPERATIONAL AND FUNCTIONAL REQUIREMENTS

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Protection Scheme.3

MODULE B OPERATIONAL AND FUNCTIONAL REQUIREMENTS

5 DISPLAY OF SYSTEM DATABASE INFORMATION

5.1 An ECDIS should be capable of accepting and converting an ENDS and their updates